SSQ-LR: Sub-FP16 Recurrent State Quantization for Hybrid Reasoners

Anonymous authors
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Abstract

This is a living paper shell for a preregistered Mac-first experiment. SSQ-LR tests whether recurrent SSM state in hybrid reasoning models can be quantized below FP16 while preserving quality. No positive result or native systems claim is made yet.

1 Current Gate

The branch must pass state-distribution heterogeneity, state-only quantization sensitivity, and cross-model transfer gates before any GPU validation. S1 must first show real state heterogeneity on a live hybrid model; S2 then freezes one state-only quantization recipe; S3 tests that recipe without per-model retuning. Failing any gate kills the branch before GPU work.

2 Mac Artifact Status

A deterministic synthetic S1 packet validates the readout format before real hybrid-model state dumps. It reports late/early max-abs ratio 8.461, std ratio 3.640, and kurtosis ratio 3.141, producing `SYNTHETIC_PASS_REAL_STATE_DUMPS_NEXT`. This is synthetic-only: it is not model evidence, not a quality result, and not GPU evidence. The metadata-only model eligibility packet identifies live hybrid targets (`granite-4.0-h-tiny`, `granite-4.0-h-small`, `granite-4.0-h-small-FP8`, and `Qwen3-Next-80B-A3B-Instruct`) and ties them to shared architecture-map hashes, but all remain blocked for real S1 on this Mac because no corresponding model weights are cached repo-locally.

3 Next Evidence

The next admissible row is a real SSM-state dump from the smallest available hybrid reasoner. Promotion requires S1 heterogeneity on real states before any state-quantization recipe is tuned.

4 Admissible Real Packet

The real S1 packet must include model and tokenizer revisions, prompt manifest hash, `prompt_ids_hash`, seed list, context lengths, dtype, device, command, and the `architecture_map_hash` from the shared config-derived maps. Rows must report `model_id`, `model_revision`, `prompt_id`, `layer`, `position_bucket`, `state_shape`, `max_abs`, `rms`, `std`, `kurtosis`, `outlier_mass`, and a `bf16_no_quant` control. The packet is valid only if every prompt/layer pair covers `prefill_end`, `2k_or_end`, `8k_or_end`, and `final_minus_128` position buckets and at least 12 prompt IDs unless `config.json` records an explicit resource-limit note. Any resource-limited packet must use a `RESOURCE_LIMITED_NOT_PROMOTABLE` decision. The summary must include the recomputed S1 evaluator fields: gate name/status/pass flag, prompt count, position buckets, SSM layer count, required and observed passing-layer counts, selected S1 ratio and CI lower bound, Holm-adjusted readout, and final-minus-128 versus prefill-end max-abs/std/kurtosis ratios. The checker recomputes these fields from raw rows and rejects stale summaries. It must pass `experimental.shared.check_gate_packet --mode real --project ssq_lr`. Saved tensor packets should be converted with `experimental.shared.hybrid_trace_packet_builder` before validation.

Required baseline	Purpose
BF16 no-op replay	Catch hook/serialization bugs.
INT8 state-only	Standard low-bit state control.
FP8-style state-only	Checks whether sub-FP16 state can match the likely deployed fallback.
MXFP4 state-only	Candidate sub-FP16 recipe.
Random same-L2 noise	Separate structure from noise magnitude.
Scale shuffle	Test scale assignment sensitivity.
Byte accounting	Include scale/metadata overhead.

Table 1: Controls required before SSQ-LR can claim a state-quantization recipe.

5 Reviewer Packet

The reviewer-facing claim boundary is tracked in `experimental/ssq_lr/paper/reviewer_pack.md`. It records that SSQ-LR is not camera-ready as a method paper until real S1–S3 evidence exists.

6 Claim Boundary

SSQ-LR is only about recurrent SSM-state precision in hybrid reasoners. It does not claim weight quantization, activation quantization, KV-cache quantization, native serving speed, or memory savings until a real state recipe and later GPU validation pass their preregistered gates.